Name_____ Date_____ Period_____

LESSON 7-7 · HOMEWORK PACKET · SUPPORTED · SESSION 1 · PAGES 1–2

Polygons and Perimeter on the Plane

Maryland Standard 6.G.A.3 · Learning target – I can find the perimeter and area of a polygon drawn on the coordinate plane.

REMEMBER

- Plot the vertices in order. Connect them to draw the polygon.
- Find each horizontal or vertical side from the coordinates.
- **Perimeter** = sum of all sides. **Area** of a rectangle = length × width.

WORKED EXAMPLE

Find the perimeter and area of the rectangle with vertices (−3, 2), (4, 2), (4, −1), (−3, −1).

Step 1 Top: 3 + 4 = 7 units. Side: 2 + 1 = 3 units.

Step 2 P = 7 + 3 + 7 + 3 = 20 units. A = 7 × 3 = 21 square units.

Answer **P = 20 units, A = 21 square units**

WORD BANK

vertex – a corner point

perimeter – distance around; add all sides

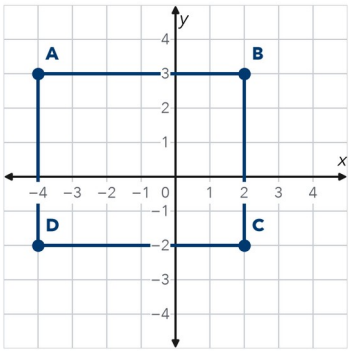
area – space inside; length × width

polygon – a closed shape with straight sides

PRACTICE

show your work · label units

1. Use the rectangle on the grid.



Step 1 Find AB: same y, so use the x-coordinates.

Step 2 Find BC: same x, so use the y-coordinates.

Step 3 Perimeter: **add** 4 sides. Area: **multiply**.

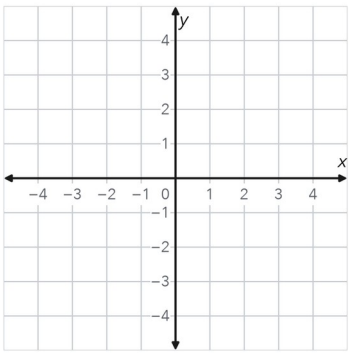
a. Length of AB _____ units

b. Length of BC _____ units

c. Perimeter _____ units

d. Area _____ sq. units

2. Plot A(−3, −2), B(4, −2), and C(−3, 4). Connect them to make a right triangle.



Step 1 Plot the three points and connect them.

Step 2 Find the base and the height.

Step 3 Area = 1/2 × base × height.

a. Base AB _____ units

b. Height AC _____ units

c. Area _____ sq. units

3. **Garden Fence.** A garden has corners at (−5, 3), (6, 3), (6, −4) and (−5, −4). Each unit is 1 meter. How much **fence** goes all the way around it?

   Step 1 Find the length of the top side.

   Step 2 Find the length of the left side.

   Step 3 Add all four sides.

   Answer_____ m

4. **Square.** A square has vertices (−3, 1), (2, 1), (2, −4), and (−3, −4). Find its perimeter.

   Answer_____ units

5. **Reasoning.** Three corners of a rectangle are (−2, 5), (3, 5), and (3, −1). Name the fourth corner.

   Frame *The fourth corner is ___. It has the same x as ___ and the same y as ___.*

   Answer_____

CHECK YOURSELF	□ I can do this on my own	□ I need a little help	□ I have a question about #___

Name_____ Date_____ Period_____

LESSON 7-7 · HOMEWORK PACKET · SUPPORTED · SESSION 2 · PAGES 3–4

Designing Polygons with Scale

Maryland Standard 6.G.A.3 · Learning target – I can use a scale to find a missing dimension of a figure on the coordinate plane.

REMEMBER

- A **scale** tells what 1 grid unit stands for, like 1 unit = 5 ft.
- Real length = grid units × scale.
- Missing side = area ÷ known side.

WORKED EXAMPLE

1 unit = 5 ft. A rectangle's sides run from (−2, 3) to (4, 3) and from (4, 3) to (4, −1). Find its real size.

Step 1 Top: 2 + 4 = 6 units. 6 × 5 = 30 ft.

Step 2 Side: 3 + 1 = 4 units. 4 × 5 = 20 ft.

Answer **30 ft by 20 ft**

WORD BANK

scale – 1 grid unit = a real length

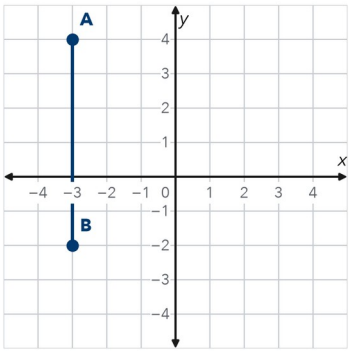
dimensions – the length and the width

area – space inside, in square units

PRACTICE

show your work · label units

1. Court Plan. On this plan, 1 unit = 5 ft. Side AB of a court is drawn. The court must cover 1,200 ft².



Step 1 Count AB in grid units.

Step 2 Multiply by 5 to get feet.

Step 3 Divide 1,200 by that length.

a. AB in grid units _____ units

b. AB in feet _____ ft

c. Other side in feet _____ ft

d. Other side in grid units _____ units

2. On a map, 1 grid unit = 4 meters.

a. 7 units = ____ m

b. 36 m = ____ units

3. Patio. A patio on a plan is 7 units by 5 units. **1 unit = 2 ft**. Find the real area.

Step 1 Change each side to feet: multiply by 2.

Step 2 Multiply the two real sides.

Answer_____ ft²

4. Bedroom. On a plan, 1 unit = 3 ft. A room has corners at (−2, 4), (2, 4), (2, −1), and (−2, −1). Find its real perimeter.

Step 1 Find each side in grid units.

Step 2 Multiply each side by 3 to get feet.

Step 3 Add all four sides.

Answer_____ ft

5. Find the Error. If 1 unit = 3 ft, Lily says (−4, 1) to (5, 1) is 3 ft. Explain her error. Fix it.

Frame *Lily should ___ because the signs are ___. The real length is ___ ft.*

_____

_____

TEST PRACTICE

choose the best answer

6. A plan uses 1 unit = 6 m. A side is 5 units long. How long is the real side?

A 11 m **B** 30 m **C** 1 m **D** 56 m

7. A rectangle has an area of 480 ft². One side is 24 ft. How long is the other side?

A 20 ft **B** 456 ft **C** 40 ft **D** 504 ft

CHECK YOURSELF	□ I can do this on my own	□ I need a little help	□ I have a question about #___